

Practical 10(Simpson 1/3 Method)

Question1:

```
In[*]:= a = Input["Enter the left end point :"];
b = Input["Enter the right end point :"];
n = Input["Enter the number of sub intervals to be formed :"];
h = (b - a) / n;
y = Table[a + i * h, {i, 1, n}];
f[x] := 1 / x;
sumodd = 0;
sumeven = 0;
For[i = 1, i < n, i += 2, sumodd += 4 * f[x] /. x -> y[[i]]];
For[i = 2, i < n, i += 2, sumeven += 2 * f[x] /. x -> y[[i]]];
Sn = (h / 3) * ((f[x] /. x -> a) + N[sumodd] + N[sumeven] + (f[x] /. x -> b));
Print["For n=", n, ",Simpson estimate is: ", Sn]
in = NIntegrate[1 / x, {x, 1, 2}];
Print["True value is ", in]
Print["Absolute error is ", Abs[Sn - in]]

For n=10,Simpson estimate is: 0.69315
True value is 0.693147
Absolute error is 3.05013×10-6
```

Question 2:

```
In[*]:= a = Input["Enter the left end point :"];
b = Input["Enter the right end point :"];
n = Input["Enter the number of sub intervals to be formed :"];
h = (b - a) / n;
y = Table[a + i * h, {i, 1, n}];
f[x] := Log[x];
sumodd = 0;
sumeven = 0;
For[i = 1, i < n, i += 2, sumodd += 4 * f[x] /. x -> y[[i]]];
For[i = 2, i < n, i += 2, sumeven += 2 * f[x] /. x -> y[[i]]];
Sn = (h / 3) * ((f[x] /. x -> a) + N[sumodd] + N[sumeven] + (f[x] /. x -> b));
Print["For n=", n, ",Simpson estimate is: ", Sn]
in = NIntegrate[Log[x], {x, 4, 5.2}];
Print["True value is ", in]
Print["Absolute error is ", Abs[Sn - in]]
```

For n=6,Simpson estimate is: 1.82785

True value is 1.82785

Absolute error is 1.50624×10^{-7}

Question 3:

```
In[*]:= a = Input["Enter the left end point :"];
b = Input["Enter the right end point :"];
n = Input["Enter the number of sub intervals to be formed :"];
h = (b - a) / n;
y = Table[a + i * h, {i, 1, n}];
f[x] := Sin[x] - Log[x] + Exp[x];
sumodd = 0;
sumeven = 0;
For[i = 1, i < n, i += 2, sumodd += 4 * f[x] /. x -> y[[i]]];
For[i = 2, i < n, i += 2, sumeven += 2 * f[x] /. x -> y[[i]]];
Sn = (h / 3) * ((f[x] /. x -> a) + N[sumodd] + N[sumeven] + (f[x] /. x -> b));
Print["For n=", n, ",Simpson estimate is: ", Sn]
in = NIntegrate[Sin[x] - Log[x] + Exp[x], {x, 0.2, 1.4}];
Print["True value is ", in]
Print["Absolute error is ", Abs[Sn - in]]

For n=12,Simpson estimate is: 4.05106
True value is 4.05095
Absolute error is 0.000109616
```

Question 4:

```
In[*]:= a = Input["Enter the left end point :"];
b = Input["Enter the right end point :"];
n = Input["Enter the number of sub intervals to be formed :"];
h = (b - a) / n;
y = Table[a + i * h, {i, 1, n}];
f[x] := Sin[x];
sumodd = 0;
sumeven = 0;
For[i = 1, i < n, i += 2, sumodd += 4 * f[x] /. x -> y[[i]]];
For[i = 2, i < n, i += 2, sumeven += 2 * f[x] /. x -> y[[i]]];
Sn = (h / 3) * ((f[x] /. x -> a) + N[sumodd] + N[sumeven] + (f[x] /. x -> b));
Print["For n=", n, ",Simpson estimate is: ", Sn]
in = NIntegrate[Sin[x], {x, 0, Pi / 2}];
Print["True value is ", in]
Print["Absolute error is ", Abs[Sn - in]]

For n=6,Simpson estimate is: 1.00003
True value is 1.
Absolute error is 0.0000263122
```

Question 5:

```

In[* ]:= a = Input["Enter the left end point :"];
b = Input["Enter the right end point :"];
n = Input["Enter the number of sub intervals to be formed :"];
h = (b - a) / n;
y = Table[a + i * h, {i, 1, n}];
f[x] := Sqrt[x] * Exp[x];
sumodd = 0;
sumeven = 0;
For[i = 1, i < n, i += 2, sumodd += 4 * f[x] /. x -> y[[i]]];
For[i = 2, i < n, i += 2, sumeven += 2 * f[x] /. x -> y[[i]]];
Sn = (h / 3) * ((f[x] /. x -> a) + N[sumodd] + N[sumeven] + (f[x] /. x -> b));
Print["For n=", n, ",Simpson estimate is: ", Sn]
in = NIntegrate[Sqrt[x] * Exp[x], {x, 1, 2}];
Print["True value is ", in]
Print["Absolute error is ", Abs[Sn - in]]

For n=12,Simpson estimate is: 5.85023

True value is 5.85023

Absolute error is  $2.95573 \times 10^{-6}$ 

```